Materials NER on unseen documents: a pre-registered evaluation on three corpora

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Abstract

MaterioMiner, a materials-science named-entity recognition (NER) corpus of four papers, is scored on random sentence splits, so its published score measures known papers, not new ones. Under protocols committed before their results, with 7 or 10 seeds and paired intervals over held-out documents, we measure the overstatement on MaterioMiner and on two corpora that hold out documents. MatSciBERT's CRF tagger scores 28 to 31 F1 points lower on a held-out paper than on random splits fine-grained (17 to 20 coarse-grained), and 17 lower on a held-out MuLMS subfield, of which 9.5 return when its training sentences come from all other subfields; on SOFC-Exp, one topic, 3.5 points are not established. Most test entities of a new paper are names no training sentence annotates (81% on MaterioMiner, 88% on MuLMS); the tagger finds a third to two fifths of them. Of 118 pre-registered decisions, 27 pass, where about 3 would by chance; allowing for shared training documents, 6 of the 18 passes paired over splits or units keep their margin. Mention replacement passes on MaterioMiner's held-out papers; four other additions are not shown to help. Four 3-seed passes failed with 10 seeds. A fine-tuned generalist tagger, GLiNER-medium, is not shown to beat the CRF tagger anywhere.

1 Introduction

Materials-science NER corpora are small. MaterioMiner (Durmaz et al., 2024) annotates 476 sentences from 4 fatigue papers with an ontology of process, structure and property classes; its reported MatSciBERT (Gupta et al., 2022) F1 is 69.92 fine-grained (27 classes) and 72.32 coarse-grained on one random 65/15/20 sentence split, averaged over 5 seeds. An extractor in use meets new papers, not new sentences of known ones, and random splits leak the papers' terms between training and test (Babl et al., 2025; Augenstein et al., 2017). SOFC-Exp (Friedrich et al., 2020; 45 articles on one topic) and MuLMS (Schrader et al., 2023; 50 articles from seven materials subfields) hold out whole articles in their official splits; we compare all three.

We ask four questions, each answered under a protocol committed before its results (for a few, part of the runs came first; Section 2; van Miltenburg et al., 2021): 1. How much is lost on a document the model has never been trained on, and where (Section 3)? 2. How far does a small-seed design mislead (Section 4)? 3. Which common additions improve materials NER on unseen documents once seeds are accounted for, and do they repeat on a second corpus (Sections 5 and 6)? 4. Does a generalist tagger do better (Section 7)?

Our runs are consistent with the published numbers. On a held-out document MatSciBERT's CRF head scores 28 to 31 F1 points lower fine-grained on MaterioMiner and 17 lower on MuLMS, of which 9.5 return when the training sentences come from all the other subfields' articles; on SOFC-Exp the 3.5 points are not established. Most of the fine-grained gap is unseen names missed outright, most of the coarse-grained gap boundaries and types (Section 3). The four 3-seed passes of stages 3 and 4 failed with 10 seeds (Section 4). Mention replacement passes on MaterioMiner's held-out papers (its votes also on MuLMS), multi-task training on its random splits; no method for unseen documents that passed on one corpus is shown to improve on the CRF head on SOFC-Exp. Of 118 pre-registered decisions, 27 pass (Section 5.3). At equal sentence counts a third annotated paper keeps about two fifths of its gain (Section 6). A fine-tuned generalist tagger, GLiNER-medium, is not shown to beat the CRF head on any corpus; given MaterioMiner's coarse types as codes, it is 10 points below there (Section 7). A companion

study (Anonymous, 2026) tests the method the programme arrived at (Section 5.2).

2 Setup

Corpora and protocols (Table 1). MaterioMiner's two levels nest: a fine-grained entity is a coarse-grained entity with the same span. SOFC-Exp's slot types refine its entity types on the same tokens. Its test articles are tokenized differently from its training articles (`cm − 2` for `cm−2`), which any model trained and tested on the released files carries (Appendix A.1). 32.9% of MuLMS's entities lie inside or on the span of another; we keep the outermost ones, so our scores cannot be compared with the published MatSciBERT-CRF (79.6 micro F1), which is scored on all of them. On MuLMS the models train on 340 sentences, about MaterioMiner's training size: for a held-out subfield, from two or three whole articles of the other subfields; for a random split, from all 50 articles.

Systems. The encoder is MatSciBERT (Gupta et al., 2022) with a token classifier on the first subword of each word; the CRF head adds a linear-chain CRF with IOB2 transition constraints (Lafferty et al., 2001; Lample et al., 2016) and is the reference of every decision from stage 5 on. A seed vote keeps the entities predicted by at least 2 of 3 seeds; multi-task training puts a fine-grained and a coarse-grained CRF head on one encoder (on SOFC-Exp, an entity head and a slot head). Section 5 tests label semantics (Ma et al., 2022), self-training (Liang et al., 2020), document context (Schweter and Akbik, 2020), document-level label propagation (Gale et al., 1992; Krishnan and Manning, 2006), mention replacement (Dai and Adel, 2020), larger encoders (He et al., 2023) and others; Appendix A.2 and A.3 give the settings. The encoders' pre-training texts may contain test articles, so "never seen" holds for fine-tuning.

Stages and seeds. A stage is one round of runs under its own protocol, committed before its results; stages 1 to 21 ran over one month (Table A1). Stages 1 and 2 used one run per split, stages 3 and 4 three seeds. From stage 5 on, every system has 10 model seeds per split (offsets 0 to 900), except the 3-seed screens and ablations, stage 8's 3-seed design, and two sets of decisions on seeds 300–900, which a 3-seed run had not used: MuLMS's and stage 20's.

Scoring and the decision rule. We score entity-level micro F1 with seqeval (Nakayama, 2018) in its default (conlleval) mode, as the dataset's code does (Tjong Kim Sang and De Meulder, 2003). A change improves on its reference only if the 95% Student-t confidence interval of the mean paired micro-F1 difference lies above 0 (a paired t test; Dror et al., 2018). The pairs are splits or held-out units, each side the mean of its seeds, or, on SOFC-Exp's official split, the seeds, an interval conditional on that split's articles. With 4 papers the interval has 3 degrees of freedom, with 5 folds 4, with 7 subfields 6, so a gain must be consistent across units to pass. It treats the units as independent, though they share training documents (Section 5.3), and with 4 or 5 units no exact sign-flip test can reach the one-sided 2.5% level, so a pass there also rests on the t distribution's assumption of normality. A pooled decision takes the units of two corpora together and requires, in addition, that neither corpus's own interval lie below 0. Each protocol states the false passes expected if nothing works (2.5% of its decisions). Square brackets give such an interval: after a difference, over the units named (or seeds or draws where stated); after a score, over its splits, units or seeds. A *screen* (3 seeds) chooses what to decide and is never a result; a *reported* comparison is named in a protocol without a decision; a *post hoc* analysis is chosen after the results, and an *exploratory* one has no protocol.

Pre-registration and runs. Each protocol was committed before its results, and before its runs but for stage 14's (pushed just after its first batch) and D5 as executed (its MuLMS means include screen runs under way when the pooled decision was added). Some sides were stored runs whose results were known: stage 5's re-tests kept stage 4's seeds, and stored runs formed the names' side of stages 14 and 18, the article pool of 20.1 and the CRF head of stage 21. Every decision from stage 5 on rests on deterministic GPU runs, and the two machines are never compared within a decision (Appendix A.8).

The count and the release. Section 5.3 counts every decision of stages 2 to 21; stage 1's comparisons have no protocol and are left out. The code, the protocols as committed before each stage, every results file and the test predictions of every GPU run behind a decision will be released on publication.

	MaterioMiner	SOFC-Exp	MuLMS
Documents	4 fatigue papers (41, 81, 201 and 153 sentences)	45 articles, one topic	50 articles, 7 subfields
Entities	fine-grained: the 27 most frequent classes (1,328); coarse-grained: 27 types (2,194)	4 entity types (4,590); 17 scored slot types (3,172)	13 types; the outermost 31,952 of 46,584
Splits	5 random 65/15/20 sentence splits; the released split (323 / 57 / 96, undocumented)	5 random 65/15/20 sentence splits; the official split, 26 / 8 / 11 articles	5 random splits of a 400-sentence pool
Held-out unit	1 paper (234–370 training sentences)	9 articles, 5 folds	1 subfield (351–2,384 test sentences)

Table 1. The corpora and their protocols.

3 The gap and its anatomy

3.1 Our runs are consistent with the published numbers

On the released split, scored like the dataset's code, plain MatSciBERT scores 70.19 [69.03, 71.35] fine-grained and 72.52 [71.88, 73.16] coarse-grained over 10 seeds. Both published numbers, 69.92 and 72.32, lie inside these intervals. Single runs range over 4.6 points fine-grained (68.0–72.7) and 3.6 coarse-grained (70.4–74.1), so a 5-seed mean of one split is a noisy reference.

3.2 The gap on three corpora

The seed vote of the CRF head scores 79.68 fine-grained and 78.63 coarse-grained over 5 random splits, but 48.81 and 61.10 on a held-out paper (Table 2). Every system with 10 seeds on both protocols scores 28–31 points lower fine-grained and 17–20 coarse-grained; no addition with runs on both protocols narrows the gap by more than a few points.

The gap is smaller where held-out documents bring fewer new names. SOFC-Exp's 45 articles share one topic and one annotation scheme. There the CRF head scores 86.27 [84.82, 87.71] over 5 random splits and 82.79 [78.33, 87.24] over 5 folds of 9 held-out articles: a gap of 3.48 points, not established (Welch 95% interval [−0.90, +7.86]; post hoc). About two of the three and a half points are boundaries (under any overlap the gap is 1.35).

On MuLMS the CRF head scores 65.72 [65.25, 66.19] over 5 random splits and 48.66 [40.91, 56.40] over the 7 held-out subfields, a gap of 17.06 points (Welch 95% interval [+9.31, +24.81]; post hoc). The gap mixes the change of topic with fewer training documents at equal size, and stage 20 split it (decision 20.1). Drawn by sentence from all the other subfields' articles instead of from two or three whole articles, the 400 pool sentences raise the CRF head to 58.17 [54.67, 61.67]: +9.51 [+1.12, +17.91] over the 7 subfields (10 seeds), every subfield gaining; with the post hoc allowance for pools that share articles, the interval is [−0.57, +19.60]. The sentence pool also differs in its subfields (all six against one to three) and its validation sentences. It lowers the share of unseen test names from 88% to 75% and raises recall on unseen names from 39.4 to 52.6. The remaining 7.55 points (Welch [+4.05, +11.05]) set an unseen subfield's documents against seen documents' sentences: topic shift and document novelty together, with the test sets' composition, not separated here.

3.3 Unseen documents are mostly unseen names

Following Augenstein et al. (2017), a gold test entity is *seen* if its exact words are an entity of the same type in the run's training sentences and *unseen* if they are no entity there; a third kind, seen with another type only, holds 0–2% of the entities (exploratory; all seeds and units pooled; Figure 1). - **The mix.** On random splits, 34% of the fine-grained test entities are unseen (coarse-grained 40%): random sentence splits put a paper's names on both sides, as Babl et al. (2025) find for other corpora. On a held-out paper, 81% are unseen (coarse-grained 79%). - **Missed outright.** Fine-grained, the CRF head misses 36% of the gold entities outright on unseen papers (10% on random splits). Coarse-grained, it misses only 6%, but 16% of the gold entities get the right span and the wrong type, and 21% a wrong boundary. - **Unseen names in a new paper are harder.** Fine-grained, the

Corpus	Held-out unit	Random splits	Held-out documents	Gap	Unseen test names, held out / random
MaterioMiner, fine-grained (CRF vote)	1 of 4 papers	79.68	48.81	30.9	81% / 34%
MaterioMiner, coarse-grained (CRF vote)	1 of 4 papers	78.63	61.10	17.5	79% / 40%
MuLMS (CRF head, 340 training sentences)	1 of 7 subfields	65.72	48.66	17.1	88% / 57%
SOFC-Exp (CRF head)	9 of 45 articles	86.27	82.79	3.5	46% / 30%

Table 2. The gap in micro F1 and the share of unseen test names (Section 3.3); MaterioMiner's rows are the CRF head's seed vote (the head alone: gaps of 31.0 and 18.4, Section 3.4).

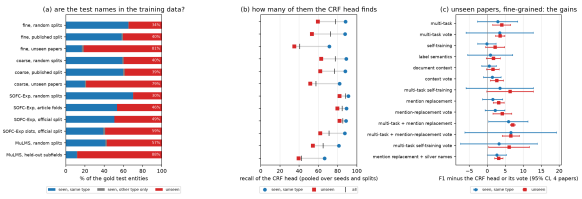


Figure 1. (a) Shares of seen, seen-with-another-type and unseen gold test entities. (b) The CRF head's recall on seen and unseen names. (c) On held-out papers, fine-grained, each method's gain over the CRF head (a vote's over the seed vote) on seen and unseen names, with 95% intervals; "silver names" are stage 11's.

CRF head finds 88% of the seen and 59% of the unseen entities of a random split, but 71% and 35% of those of a held-out paper (coarse-grained, 89% and 63% against 82% and 52%). Its fine-grained recall falls from 77.8 to 40.6; the mix and the lower recall within each kind each account for 40–60% of that gap, in either order of the decomposition. - **The other corpora.** 49% of SOFC-Exp's official test entities are unseen, and the CRF head finds 82% of them. On MuLMS's held-out subfields it finds 39% of the unseen names (66% of the seen), and their larger share accounts for about a third of its 22.5-point recall gap. - **The gains show on unseen names.** Paired over the 4 held-out papers (fine-grained), every gain whose interval clears 0 overall also clears it on unseen names, but for six-copy mention replacement (+2.98 [+0.09, +5.86] overall, +3.65 [−0.15, +7.46] unseen). Seen names are about a sixth of a new paper's entities, so their intervals are wide: only mention replacement on the multi-task model (+5.88 [+0.42, +11.34]) and stage 11's names (+2.66 [+0.03, +5.29]) clear 0 there.

A method for unseen documents therefore has to raise F1 on names the training documents do not contain.

3.4 Found, but cut differently

seqeval counts a prediction only if its span and type are exactly those of a gold entity. Under the looser criteria of SemEval-2013 Task 9.1 (Segura-Bedmar et al., 2013), each gold entity matched at most once (exploratory), the CRF head's coarse-grained gap of 18.4 points shrinks to 9.7 when only the span must match and to 2.6 under any overlap: on a new paper the model finds almost every entity, and loses mostly on where the boundaries go and which type applies. Fine-grained, the gap shrinks from 31.0 to 25.0 and 18.3; there the model misses entities outright. On held-out papers, 31% of the fine-grained predictions overlap a gold entity but differ in span or type (13% on random splits). The gold annotations differ too: only 35 fine-grained names are entities in two or more papers, and 7 of them take different types. Part of what looks like a failure to generalize may be a difference in annotation conventions between papers (a hypothesis; we did not adjudicate the annotations).

3.5 A uniform decoding bias does not help

On unseen papers the CRF head predicts too few entities. A natural repair adds a bias D to the emission score of every tag but O before Viterbi decoding, trading precision for recall (Lipton et al., 2014; Minkov et al., 2006); we chose D per held-out paper on inner folds, without the paper's labels, and per run on SOFC-Exp's validation articles. It is not shown to help: −0.59 [−1.75, +0.57] fine-grained and +0.17 [−0.14, +0.48] coarse-grained on the 4 held-out papers, −0.21 [−0.58, +0.17] on SOFC-Exp. The missed entities are not near the threshold: pooled over papers and seeds, the added predictions are right 22% of the time fine-grained (D = +0.25), 19% coarse-grained and 21% on SOFC-Exp, below the break-even rate for F1, F1/2 (23%, 29% and

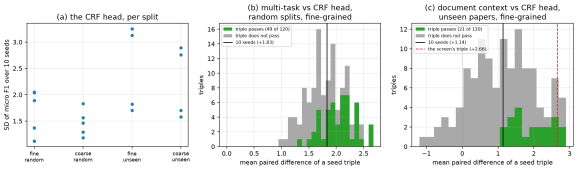


Figure 2. (a) The CRF head's SD over 10 seeds, per split. (b, c) The mean differences of the 120 seed triples, fine-grained, for multi-task training on random splits and document context on unseen papers, against the 10-seed difference.

42%).

4 Seeds and machines

One split's micro F1 varies over the 10 seeds with a mean SD of 1.5–1.7 points on random splits and 2.2–2.5 on unseen papers (Figure 2a). The seed vote, a three-model ensemble reported rather than decided, gains 1.67 to 2.50 points over the mean single CRF run on MaterioMiner, every interval above 0, and also on MuLMS (50.54 against 48.66) and on SOFC-Exp's official split (85.13 against 83.75). From the 120 seed triples of the 10-seed runs, the 1.83-point multi-task gain passes in 41% of the triples (49 of 120), and up to 37% of triples give the sign opposite to the 10-seed difference; that pass itself loses its margin under the corrected resampled test (Section 5.3).

Of the 22 decisions of stages 3 and 4, taken with 3 seeds, four passed, three of them on CPUs (Table B1). Re-decided with 10 seeds on GPUs in stage 5, none passed. For the three CPU passes the re-test changed the machine as well as the seeds, and the GPU runs of the same seeds 0, 100 and 200 already fail (+0.07, −0.08 and +0.40, every interval covering 0; post hoc). The same configuration and seed differ between the two machines by a median of 1.43 F1 points over 186 matched runs, and no pair is identical (post hoc): a 3-seed interval on one machine misses run-to-run variation that another machine, or more seeds, exposes.

The stage-6 screen shows the same: document context passed it with +2.66 [+0.91, +4.40] fine-grained, and on the 7 seeds the screen did not use it gains +0.49 [−3.10, +4.07] (post hoc); only 21 of its 120 triples pass. Screens then only chose what to run with 10 seeds, though the decisions of stages 6 and 7 still averaged in the screen's three seeds.

5 The additions, tested

Each addition is decided against the CRF head on held-out documents, and its vote against the strongest system on unseen papers; a methods claim also needs its gain to repeat on a second corpus. Tables B1 to B3 list the decisions.

5.1 What holds, and where

The CRF head gains +3.12 and +2.96 over plain MatSciBERT on the published split; under strict IOB2 scoring the gain shrinks to +1.70 [+0.21, +3.19] and +1.07 [−0.82, +2.97], so much of it comes from predicting only valid spans. On SOFC-Exp it gains +1.22 [+0.50, +1.93]. On test articles cut into words as the training articles are, its 84.61 [83.89, 85.34] covers the published MatSciBERT-CRF score (84.77; Gupta et al., 2022), which its 83.75 [82.85, 84.65] on the released test files does not, and its gain falls to +0.17 [−0.37, +0.70]: it mostly repairs the tokenization shift, whose two measurements pass (+0.86 [+0.39, +1.33] for the CRF head, +1.91 [+1.33, +2.49] for plain MatSciBERT).

Multi-task training is the one training change that holds on MaterioMiner's random splits, +1.83 [+0.32, +3.34] fine-grained (coarse-grained +0.71 [−0.15, +1.56], no), and its vote improves on the seed vote coarse-grained there (+0.79 [+0.17, +1.42]) and fine-grained on unseen papers (+3.65 [+0.49, +6.82]). On SOFC-Exp's two levels it is not shown to gain (−0.78 [−2.39, +0.84] on the slot types, −0.15 [−1.36, +1.05] on the entity types).

Mention replacement (Dai and Adel, 2020) makes three copies of each training sentence with an entity, each mention replaced with probability 0.5 by a training mention of the same type. Screened with 3 seeds and decided with 10 only because the screen's mean was positive (+3.25 [−2.33, +8.82]), it improves on the CRF head on MaterioMiner by +2.77 [+1.51, +4.03] fine-grained and +1.28 [+0.06, +2.51] coarse-grained, every held-out paper gaining (the decision averages in the screen's three seeds; on the other seven, post hoc, +2.57 [+0.10, +5.03] and +1.73 [−0.63, +4.08]). On SOFC-Exp it does not pass, on the official split (−0.29 [−1.04, +0.47]), over the folds (−0.48 [−1.36, +0.39]) or at MaterioMiner's size (+0.50 [−0.47, +1.46]). On MuLMS the single model gains +2.10 [−0.31, +4.51] and its votes +2.71 [+0.22, +5.20], which passes. By the pre-registered criterion, it stays a MaterioMiner result.

On the multi-task model, mention replacement gains, on MaterioMiner, +3.01 [+0.42, +5.60] fine-grained and +2.30 [+0.49, +4.10] coarse-grained over the multi-task model and +6.73 [+6.03, +7.43] fine-grained over the CRF head, the sum of its parts; its vote gains +2.86 [+0.18, +5.54] over the multi-task vote. It is not shown to gain on SOFC-Exp. 3 of its 7 decisions pass, where about 0.18 would by chance: a result for training sets of a few papers with two type levels.

Document-level label propagation (Appendix A.3) passes coarse-grained on MaterioMiner (+0.12 [+0.01, +0.24]) and on MuLMS (+0.82 [+0.18, +1.46], 13.1a), and loses on SOFC-Exp's official split (−0.42 [−0.61, −0.22]), whose annotators leave 14% of a term's repeated occurrences untagged. It is a two-corpus claim only under the criterion revised in stage 13 (stage 7's protocol required SOFC-Exp), and the effect is under a point.

Not shown to help with 10 seeds: label semantics (+1.23 [−0.96, +3.42] fine-grained on unseen papers), self-training (+1.74 [−1.16, +4.64]), document context (+1.14 [−0.94, +3.22]), multi-task self-training (+5.47 [−0.74, +11.68]; three papers gain 4–9 points, and the screen's gain held on its unused seeds, post hoc: a possible gain the design cannot confirm), recall calibration and coarse-to-fine completion (+0.34 [−0.91, +1.59]). DeBERTa-v3-large's 3-seed screen, at MatSciBERT's untuned learning rate, did not pass (−4.4 and −8.0 on unseen papers).

5.2 Names from unlabeled text

The companion study (Anonymous, 2026) harvests the names a new document brings: three CRF teachers tag 40,000 unlabeled sentences on the held-out documents' topic, and the mentions all three tag alike, every marginal at least 0.9, enter training through mention replacement with six copies (MR6; with three, MR3). Of its twelve decisions (Table B2), four pass: two on MaterioMiner (+4.27 [+1.39, +7.15] fine-grained), a pooled one (+3.58 [+0.95, +6.21]) whose criterion was adopted when MR3 already met it, and recall on the unseen names the lexicon holds (+11.24 [+3.31, +19.18]), which passes only with three subfields no decoding bias matched and whose outcome was chosen after a post hoc split had found the effect. On the pooled decision's units and seeds MR3 alone gains +2.35 [+0.94, +3.75]; on the 11 units with MuLMS's decision seeds the names add +0.80 [−0.66, +2.26] over MR6 (D7). A lexicon ranked by a graph network failed its gate (stage 19; Table A1).

5.3 The count

Through stage 21 the programme took 118 preregistered decisions (stages 15, 16 and 19 took none), 15 three-seed screens (8 in Table B1, counted per granularity, and the companion study's 7) and stage 19's lexicon screen, which failed. 27 decisions pass, where about 3.0 would if nothing worked: none of the 6 with one run per split; 6 of the 24 with 3 seeds (4 of the 22 of stages 3 and 4, none of which survived its 10-seed re-run, and stage 8's 2 on a third paper, re-run in stage 20); and 21 of the 88 with 7 or 10 seeds (Table 3), none of them stage 21's.

Three post hoc readings qualify the 27 passes; none changes a verdict. Stage 20's three passes carry the reading fixed in their protocol (the coarse-grained third paper keeps its margin, the other two lose it); stage 21 has none, and 21.2 stays below. - **Shared training documents.** The companion study's model of the units' dependence (Anonymous, 2026; Appendix A.5) and, on random splits, the corrected resampled test (Nadeau and Bengio, 2003) widen the intervals by a factor of 1.2 to 1.6: of the 18 passes over units with 7 or 10 seeds, 6 keep their margin (Table 3). - **Multiplicity.** About 3.0 of the 27 passes are expected by chance, a false-discovery share near 11%. Under Benjamini–Hochberg control of the 118 one-sided P values (Benjamini and Hochberg, 1995), 3 pass at a 5% false-discovery rate (SOFC-Exp's two tokenization measurements and stage 3's CPU label-semantics pass, which did not replicate) and all 27 at 10%, a fragile count: the largest of their P values, 0.0223, falls just under its step-up threshold, 0.0229 (over stages 2 to 18's 111 decisions, 12 passed). Within each addition's family (Holm, family-wise 2.5%), 9 survive, stage 20's three measurements among them, none a method for unseen documents with 7 or 10 seeds. - **Power.** At the variance in their own data, the single-model decisions on held-out papers could detect 0.4 to 3.9 points at 80% power (multi-task self-training's fine-grained one 8.1, GLiNER-medium's 12.0 and 7.3). Each failure is still read by its interval: "not shown", with the upper bound as the largest gain the data allow (Tables B1 to B3), not "no effect". - **Deviations.** Not every decision is confirmatory in the strict sense: the pooled names decision's criterion was adopted when three-copy

Addition	Decisions	Pass	Where it holds	Margin kept with the dependence allowance (post hoc)	Passes BH at 5% (post hoc)
CRF head over plain MatSciBERT	1 (SOFC-Exp's official split)	1	MaterioMiner (reported) and SOFC-Exp, mostly a tokenization repair	paired by seed	no
seed vote	none (reported)	—	adds 0.8 to 2.5 points wherever measured	—	—
multi-task training	6 on MaterioMiner (4 on random splits), 2 on SOFC-Exp	3 / 0	MaterioMiner only	none	none
mention replacement	4 on MaterioMiner, 2 on MuLMS, 3 on SOFC-Exp	2 / 1 (votes) / 0	MaterioMiner; MuLMS by vote	fine-grained MaterioMiner	none
mention replacement on the multi-task model	4 on MaterioMiner, 3 on SOFC-Exp	3 / 0	MaterioMiner only	none	none
harvested names (stage 11; D1 to D12)	16: 7 on MuLMS, 6 on MaterioMiner, 2 pooled, 1 on SOFC-Exp	4 (D3, D4, D5, D11)	MaterioMiner, and pooled with MuLMS under a criterion adopted mid-study; on MuLMS alone only on recall, with three subfields unmatched; not over MR6 in F1; not SOFC-Exp	D4, D5, D11	none
label propagation	10: 6 on MaterioMiner, 2 on MuLMS, 2 on SOFC-Exp	2	MuLMS and coarse-grained MaterioMiner; loses on SOFC-Exp	MuLMS (13.1a)	none
label semantics, self-training, document context, multi-task self-training, recall calibration, coarse-to-fine	28, 2 of them on SOFC-Exp	0	nowhere	—	—
SOFC-Exp's test articles re-cut (a measurement)	2	2	—	paired by seed	both
MuLMS's pool by sentence; a third paper at equal size, fresh seeds (measurements, stage 20)	3	3	MuLMS; MaterioMiner at both levels	the coarse-grained third paper (fixed in the protocol)	none
GLiNER-medium fine-tuned (stage 21)	4: 2 on MaterioMiner, 1 on MuLMS, 1 on SOFC-Exp	0	nowhere above the CRF head; below it at MaterioMiner's coarse level, whose types it was given as codes	no pass; 21.2 stays below	none

Table 3. Decisions with 7 or 10 seeds, by addition (8 on MaterioMiner's random splits). The two right-hand columns are post hoc; stages 20 and 21 fixed their dependence reading in their protocols, and the post hoc model reproduces it.

mention replacement already met it, and three decisions would not have run under the original one; the recall decision's outcome followed a post hoc split; label propagation's criterion was revised in stage 13; stages 6 and 7 averaged in the screen's seeds; stage 14's protocol came just after its first batch; and stage 5's re-tests reused known seeds (Section 2, Table B2).

6 Data against method

The inner folds of Section 3.5 train the CRF head on two papers. On each held-out paper, training on all three other papers instead adds +7.36 [+1.98, +12.74] F1 fine-grained and +8.90 [+5.90, +11.90] coarse-grained, over 4 papers and seeds 0–200 (exploratory); every paper gains. A pre-registered run trains on three papers cut to the size of two. At the same number of sentences the third paper adds +1.18 [+0.22, +2.15] fine-grained and +3.59 [+1.59, +5.58] coarse-grained on stage 8's 3 seeds. Stage

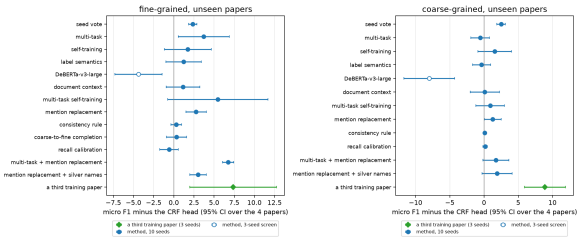


Figure 3. Each method's micro-F1 difference from the CRF head on the 4 held-out papers, with 95% intervals: 10 seeds, or a 3-seed screen (hollow), and last a third training paper (3 seeds, exploratory).

20 re-decided both on seeds 300–900, which stage 8 had not seen (decisions 20.2 and 20.3): at equal size the third paper adds +4.03 [+1.06, +7.00] fine-grained and +3.52 [+1.32, +5.72] coarse-grained, again on every paper, and its own sentences +6.19 [+0.21, +12.17] and +5.10 [+3.57, +6.64]. With the post hoc allowance for shared papers, the fine-grained interval reaches −0.51 and the coarse-grained one keeps its margin ([+0.16, +6.88]). By point estimate, diversity is about two fifths of the gain at both levels on the fresh seeds (a sixth fine-grained on stage 8's 3).

On top of two papers, a third adds about as much as the largest method gains measured on top of three (mention replacement on the multi-task model, +6.73 fine-grained, Figure 3; with six copies and harvested names, +7.49 [+5.96, +9.02], reported in stage 17). The baselines differ, the third paper's 3-seed gain is exploratory, and annotation cost is not weighed.

7 A generalist baseline

Stage 21 tests a generalist tagger, which need not depend on the training documents' names: GLiNER-medium v2.1 (Zaratiana et al., 2024), an open-weight span model pre-trained to tag any label set it is given by name, on a base-size encoder (DeBERTa-v3-base), chosen because it fits a free T4 GPU. Its protocol was fixed before any run, after every held-out result of the CRF head was known.

A zero-shot arm runs once per unit on MaterioMiner fine-grained, MuLMS and SOFC-Exp's folds, its threshold chosen on validation micro F1; a fine-tuned arm trains on the CRF head's training sentences, splits and 10 seeds, with a fixed recipe and threshold 0.5. The labels are the types' names in words (e.g. "stress amplitude"; Appendix A.7). MaterioMiner's coarse-grained types entered as their codes lowercased (e.g. "mec"), with no zero-shot arm, because the protocol took the codes to be undocumented; the dataset's release names every code, which we found after the run.

Decisions 21.1–21.4 take the fine-tuned model minus the CRF head, each unit the mean of its 10 seeds, paired over MaterioMiner's papers fine-grained (21.1) and coarse-grained (21.2), MuLMS's subfields (21.3) and SOFC-Exp's folds (21.4), with about 0.1 false passes expected. The protocol expected a null to bound effects of about 4–5 points on MaterioMiner, 3.5 on MuLMS and 1.5–2 on SOFC-Exp; we read each by its interval. MaterioMiner counts as one corpus, above or below the CRF head only if both its levels are. The zero-shot arm, recall by kind of name and a dependence allowance fixed in advance are reported without a decision. Every reading carries GLiNER's differences from the CRF head together: generalist NER pre-training, a general-domain encoder, a span classifier with its loss and label conditioning, and its tokenizer.

None passes, and the intervals are wider than the protocol expected: they allow −5.4 to +12.9 points on MaterioMiner's fine-grained types (one paper +12.3, the other three −0.5 to +1.7), −1.9 to +5.1 on MuLMS and −1.8 to +2.4 on SOFC-Exp. With its coarse types given as codes, GLiNER is 10.1 points below on every paper and stays below with the allowance; its F1 is lower coarse-grained than fine-grained, while the CRF head's is 12 points higher (exploratory); given class names, it loses less (below). Zero-shot, it trails by 20 to 48 points. Its recall on unseen names is higher only on SOFC-Exp, where its F1 is not shown to differ; on MaterioMiner fine-grained and MuLMS no difference is shown, and coarse-grained it is lower (no multiplicity control). Its validation F1 is lower in every setting, and its recipe was fixed, not tuned, with our own training loop (Appendix A.7).

Two exploratory checks, fixed before their runs and outside the count, follow up the codes and the loop. Given the coarse types' class names (e.g. "mechanism" for "mec"), the fine-tuned model scores +6.16 [−0.35, +12.67] more, higher on every paper, and its gap to the CRF head narrows to −3.97 [−9.15, +1.22] (Table 4); 21.2 stands as taken. The package's own trainer, with the same recipe and stopping epochs, differs from our loop by −0.25 [−3.93, +3.42] over 10 paired seeds on one paper.

Setting (decision)	CRF head	Zero-shot − CRF head	Fine-tuned; − CRF head	Validation F1, fine-tuned / CRF head	Unseen-name recall, fine-tuned − CRF head
MaterioMiner fine (21.1)	46.46	−19.87 [−30.73, −9.00]	50.23; +3.77 [−5.38, +12.92]	74.15 / 79.47	+7.66 [−5.20, +20.52]
MaterioMiner coarse (21.2)	58.60	no arm	48.48; −10.12 [−15.68, −4.56]	70.24 / 79.33	−17.01 [−23.47, −10.56]
MaterioMiner coarse, class names (exploratory)	58.60	no arm	54.64; −3.97 [−9.15, +1.22]	74.44 / 79.33	not computed
MuLMS (21.3)	48.66	−32.44 [−38.71, −26.17]	50.23; +1.58 [−1.92, +5.08]	74.03 / 76.89	+2.68 [−1.19, +6.54]
SOFC-Exp folds (21.4)	82.79	−47.78 [−48.85, −46.70]	83.05; +0.27 [−1.84, +2.37]	85.62 / 86.44	+1.89 [+0.54, +3.25]

Table 4. Stage 21, GLiNER-medium v2.1 against the CRF head: micro F1 means over the units (zero-shot one run, fine-tuned 10 seeds), the decisions and the reported comparisons, with 95% intervals over the units. The exploratory row gives the coarse types' class names in place of their codes.

8 Lessons

1. **Hold out documents, and say which.** A random sentence split scores the reading of known papers, a held-out paper or subfield what an extractor will meet. Report both, with the share of unseen test names: the gap is largest where held-out documents add the most (+47, +39, +31 and +17 points over random splits, against gaps of 31, 17.5, 17 and 3.5), an ordering over four settings rather than a predictor.
2. **Count unseen names and match criteria separately.** A single strict F1 hides whether the gap is missed names or boundaries and types.
3. **More held-out units, and ten seeds.** Three seeds pass a 1.8-point gain that ten seeds pass only 41% of the time, and passed four comparisons that ten seeds on GPUs did not confirm. With four or five units, the units, not the seeds, set an interval's width. A 3-seed screen chooses what to run, on seeds the decision will not use; it is never a result.
4. **Take the vote.** Three seeds' majority adds 0.8 to 2.5 points wherever measured (reported, not decided).
5. **Do not assume a method transfers.** Multi-task training, mention replacement and their combination each passed on MaterioMiner and were not shown to gain on SOFC-Exp, whose held-out articles bring fewer new names and whose baseline leaves less headroom; which of the two blocks them is not known.
6. **Pre-register and count.** Of 118 decisions, 27 pass; a paper that reported the 27 alone would read as a different programme.
7. **Annotate new documents.** On top of two papers, a third adds about as much as the best method adds on top of three (Section 6).

9 Related work

Materials NER. Weston et al. (Weston et al., 2019) trained an NER model on annotated materials abstracts and applied it to millions of abstracts. MatSciBERT (Gupta et al., 2022) is pre-trained on materials papers, Trewartha et al. (Trewartha et al., 2022) compare general, scientific and materials pre-training, and MatSci-NLP (Song et al., 2023) gathers materials NLP tasks into one benchmark. Potu et al. (Potu et al., 2025) study a shift between two fatigue datasets; we study the shift to an unseen paper of the same dataset. Large language models now extract materials data by fine-tuning (Dagdelen et al., 2024) or by prompting (Polak and Morgan, 2024); we evaluate fine-tuned encoder taggers and one generalist tagger (Zaratiana et al., 2024).

Splits and unseen mentions. Whether random splits suffice is contested (Gorman and Bedrick, 2019; Søgaard et al., 2021). For NER, random splits leak entity names into test and inflate scores (Babl et al., 2025), and performance is far lower on mentions unseen in training (Augenstein et al.,

2017), where contextual embeddings help most (Taillé et al., 2020); coverage of test entities by training explains much of a score (Fu et al., 2020; Lin et al., 2020), models lean on the regularity of names (Ghaddar et al., 2021) and fail on swapped ones (Agarwal et al., 2020), biomedical taggers generalize poorly to novel entities (Kim and Kang, 2022), and cross-domain benchmarks make the shift explicit (Liu et al., 2021). We measure both effects by leaving whole documents out.

Seeds, significance and pre-registration. Seed variance can mislead comparisons of single runs (Reimers and Gurevych, 2017; Dodge et al., 2020; Bouthillier et al., 2021). Few units leave decisions with little power (Card et al., 2020), units that share training data make intervals too narrow (Nadeau and Bengio, 2003), and many decisions call for a false-discovery count (Benjamini and Hochberg, 1995) or a replicability analysis across datasets (Dror et al., 2017); Section 5.3 reports the first three, and Table 3's corpus-by-corpus verdicts stand in for the last.

Methods for unseen documents. We did not test generative augmentation (Ding et al., 2020; Zhou et al., 2022). Mention replacement's gains for transformers shrink as training data grows (Dai and Adel, 2020). Du et al. (Du et al., 2021) retrieve unlabeled sentences for self-training from the web; we use an in-domain corpus of open-access papers. CRFs can be trained for F-score directly (Suzuki et al., 2006), and calibration degrades under dataset shift (Ovadia et al., 2019), which Section 3.5 finds. Harvesting names from unlabeled text, by bootstrapping (Kozareva, 2006) or teacher–student training (Wu et al., 2020; Qu et al., 2023), is the companion study's subject.

10 Limitations

- **Encoder taggers.** Every system is a fine-tuned BERT-base tagger, but for DeBERTa-v3-large's screen and stage 21's generalist tagger, run at base size with a fixed recipe (Section 7). No large language model was run on the held-out documents; whether one narrows the gap is open.
- **Few units.** Most method results rest on MaterioMiner's four papers, a case study; with four or five units every pass also rests on the t distribution's normality (Section 2).
- **MuLMS's small-data regime.** Its models train on 340 of its 10,184 sentences, to match MaterioMiner's size; its gap describes that regime.
- **Model selection inside the training documents.** The epoch is chosen on a random 15% of the training documents' own sentences (on MuLMS, 60 sentences of the pool), the kind of split this paper argues against.
- **SOFC-Exp's methods on one split.** Most are compared on its official split, with intervals over seeds only; mention replacement, its multi-task combination and the companion study's method are also decided on the folds.
- **Methods tested on MaterioMiner only.** Self-training (with a fatigue corpus) and coarse-to-fine completion rest on MaterioMiner alone (on MuLMS self-training was only screened with 3 seeds), as does recall calibration beyond one SOFC-Exp split.
- **The released split** is undocumented. Its use reproduces the published numbers, but the dataset's authors have yet to confirm that it is theirs.
- **Annotation conventions differ between papers** (Section 3.4). The decisions use strict F1 for comparability; the looser criteria are reported, not decided on.
- **The count is of this programme's choices.** The 118 decisions test the additions we chose, in the order we chose them. About 3.0 would pass by chance whatever their dependence, which only makes chance passes come together; but the votes share members, pooled decisions share units with per-corpus ones, and units share training documents, which makes the intervals over units too narrow, so the chance count is a lower bound (Section 5.3).

Use of an AI assistant

An AI assistant, Claude Code (Anthropic), was used under the authors' direction to write the code, run the experiments, draft the text and review the drafts; the authors took the study's decisions and are responsible for the content.

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A Replication material

A.1 Corpora and preprocessing

MaterioMiner (Durmaz et al., 2024): 476 sentences, 12.2k tokens, from 4 papers of 41, 81, 201 and 153 sentences. - Fine-grained, the 27 most frequent classes are kept, as in the dataset paper (ties broken by token count, then name). They hold 1,328 of the 2,189 fine-grained entities. Coarse-grained, all 27 types are kept: 2,194 entities. - The

two levels nest: a fine-grained entity is a coarse-grained entity with the same span, and each fine-grained class has one coarse-grained parent. 15 sentences are tokenized differently in the two files. - The release ships a split of 323 / 57 / 96 sentences. The dataset paper does not describe it, and it partitions the two granularities differently. - Random splits: five random 65/15/20 sentence splits, one per split seed. Held-out papers: train on three papers, with 15% of their sentences for validation, and test on the fourth.

SOFC-Exp (Friedrich et al., 2020): 876 annotated sentences from 45 articles on solid oxide fuel cells, 35.0k tokens, 4,590 entities of 4 types (MATERIAL, VALUE, DEVICE, EXPERIMENT). - The official split holds out whole articles: 26 / 8 / 11 articles, 568 / 135 / 173 sentences. The folds hold out 9 articles each, five folds. - Slot types refine the entity types on the same tokens (e.g. anode material or working temperature); each falls within one entity type. The 17 that the dataset's evaluation scores hold 3,172 mentions (702 in the test articles). - Tokenization. ChemDataExtractor's tokenizer (Swain and Cole, 2016) reproduces the tokens of all 7,630 sentences of the 34 training and development articles; Java's word BreakIterator reproduces 1,833 of the 1,836 sentences of the 11 test articles. A model trained on `cm−2`, `O2−` and `high - temperature` is tested on `cm − 2`, `O2 −` and `high-temperature`. The re-cut test articles of Section 5.1 are the test articles cut as the training articles are.

MuLMS (Schrader et al., 2023): 50 articles from seven materials subfields (in the release: electrolysis, graphene, pemfc, poly, semi, sofc and steel; 2–11 articles each), 10,184 sentences and 46,584 entities of 13 types (CC BY-SA 4.0). - 32.9% of the entities lie inside or on the span of another. A flat tagger needs one level, so we keep the outermost ones: 31,952 (68.6%). Of entities with the same span we keep the type first in a fixed order (the larger unit first: VALUE over NUM); of two that cross, the one that starts first. Most kept entities are materials (12,647) and values (4,628). - Text extraction damaged 276 tokens: 261 are empty or hold whitespace, and 15 are private-use symbols or lone combining marks, which a BERT tokenizer erases. Each token becomes its whitespace-separated pieces, without the pieces a BERT tokenizer erases; 2 entities are left without tokens and dropped. - The published MatSciBERT-CRF (79.6 micro F1) tags the nested entities with stacked labels and is scored on all of them. - Held-out subfields: each of the 7 subfields is held out in turn, the model trained on 340 sentences and validated on 60 of a 400-sentence pool from two or three articles of the other subfields (a different set per held-out subfield, the articles taken whole), and tested on every sentence of the held-out subfield (351–2,384). Random splits: five random splits of a 400-sentence pool drawn from all 50 articles. Stage 20's sentence pool draws its 400 sentences per held-out subfield by sentence from all the other subfields' articles (39 to 48 articles from all six).

A.2 Systems and training settings

- **Encoder.** MatSciBERT (Gupta et al., 2022) with a token classifier on the first sub-word of each word. AdamW at 5e-5 with linear warm-up, batch size 16, up to 20 epochs, early stopping on validation F1 (patience 3).
- **CRF head.** A linear-chain CRF over the words with IOB2 transition constraints, trained by likelihood and decoded with Viterbi (Lafferty et al., 2001; Lample et al., 2016). It is the reference of every decision from stage 5 on.
- **Seed vote.** The entities predicted by at least 2 of 3 seeds; its score is the mean over the 3 disjoint seed triples (0–200, 300–500, 600–800).
- **Multi-task.** One encoder with a fine-grained and a coarse-grained CRF head, trained on the sum of their losses; on SOFC-Exp, an entity head and a slot head.
- **Seeds.** From stage 5 on, 10 model seeds per split (seed offsets 0 to 900), except the 3-seed screens and ablations, stage 8's 3-seed design, and the MuLMS decisions on the seven seeds a screen did not see (300–900). From stage 12 on, MuLMS's decisions on a screened method use only those seven seeds, but for the pooled D5; 13.1a and 13.1b, a rule applied to saved predictions, use all ten.
- **Scoring.** seqeval (Nakayama, 2018) entity-level micro F1 in its default (conlleval) mode, as the dataset's code scores it (Tjong Kim Sang and De Meulder, 2003). Strict IOB2 scoring (an I- tag must continue an entity of its type, or the span does not count), macro and support-weighted F1 are reported alongside.

- **Tested additions.** Class weights; domain-adaptive and in-domain pre-training; other encoders, SciBERT, PubMedBERT and DeBERTa-v3-large (He et al., 2023; 435M parameters against MatSciBERT's 110M, screened at MatSciBERT's learning rate; it did not fit a 15 GB GPU on SOFC-Exp at the protocol's batch size, so its planned SOFC-Exp comparison did not run); label semantics (Ma et al., 2022); self-training with 3 teachers (Liang et al., 2020), alone and on the multi-task model, on an in-domain corpus of open-access papers, one per dataset, without the dataset's own papers or any sentence that shares a word 8-gram with them; document context (Schweter and Akbik, 2020); document-level label propagation (Gale et al., 1992; Krishnan and Manning, 2006); coarse-to-fine completion; recall calibration; mention replacement (Dai and Adel, 2020), alone, on the multi-task model, and with harvested names (Anonymous, 2026).

A.3 Procedures

Mention replacement (MR3; MR6 writes six copies):

```
for each training sentence s with at
 ↪  least one entity:
    repeat 3 times:
        s' = a copy of s
        for each entity mention m in
         ↪  s':
            with probability 0.5,
             ↪  replace m by a
             ↪  training mention
            of the same type
        add s' to the training data
```

On the multi-task model, a copy replaces a coarse-grained mention, in both heads' tags at once, by a training mention of the same coarse- and fine-grained type, so both levels' tags stay valid.

Harvested names (the companion study's method; Anonymous, 2026):

```
teachers = the CRF head's runs of
 ↪  seeds 0, 100 and 200
corpus   = 40,000 unlabeled
 ↪  sentences on the held-out
 ↪  documents' topic
lexicon  = the mentions all three
 ↪  teachers tag in corpus with the
 ↪  same
           span and type, every
            ↪  marginal at least 0.9
train by mention replacement with
 ↪  six copies, the lexicon entering
 ↪  it
```

Document-level label propagation, applied to a model's saved predictions:

```
for each document:
    for each term the model tags in
     ↪  most of its occurrences:
        tag the term in its untagged
         ↪  occurrences
```

Recall calibration:

```
add a bias D to the emission score
 ↪  of every tag but O; decode with
 ↪  Viterbi
choose D per held-out paper on inner
 ↪  folds (the CRF head trained on
 ↪  two of
    the three training papers),
     ↪  without the held-out paper's
     ↪  labels;
    on SOFC-Exp, per run on the
     ↪  validation articles
```

The inner folds chose D = +1.25, +1.25, −1 and +0.25 fine-grained. Coarse-to-fine completion uses the coarse-grained level to complete the fine-grained one.

The stage-19 screen: a relational graph network (R-GCN; Schlichtkrull et al., 2018) over every name the three teachers tag in the unlabeled corpora, its relations sentence co-occurrence, character similarity and a shared head word and its node features the teachers' vote among others, was trained to type each unit's 82 to 253 training names and ranked the new names by type probability. Its rivals at equal lexicon size were the teachers' vote at threshold 0.7 (3 of 3) and the same network without the graph (a multilayer perceptron). A test-set gate over D7's 11 units decided whether the F1 decisions would run: the graph's lexicon had to hold 5 points more of the unseen test mentions than the vote and 2 more than the network without the graph. The probe keeps one type per name and only names no training sentence annotates.

A.4 The decision rule and the statuses

- **Intervals.** A change improves on its reference only if the 95% Student-t interval of the mean paired micro-F1 difference lies above 0. The pairs are splits (5) or held-out units, papers

(4; 3 degrees of freedom), folds (5; 4) or subfields (7; 6), each side the mean of its seeds; on SOFC-Exp's official split the pairs are the seeds, and the 10-article runs pair by draw.

- **Pooled decisions** take the units of two corpora together and require, in addition, that neither corpus's own interval lie below 0.
- **Votes** are compared with the strongest system on unseen papers: the multi-task vote fine-grained, the seed vote coarse-grained.
- **Expected false passes**: 2.5% of the decisions, if nothing works.
- **Statuses.** A *decision* is a comparison its protocol decides by the rule, fixed before its results. A *screen* is a 3-seed comparison that chooses what to decide, never a result. A *reported* comparison is named in a protocol without a decision. A *pre-specified* analysis is named in a protocol before its results, a *post hoc* one is chosen after them, an *exploratory* one has no protocol, and a *descriptive* number carries no inference.
- **Gaps between protocols** carry Welch 95% intervals (post hoc).
- **Seen and unseen names** (Augenstein et al., 2017). A gold test entity is seen if its exact words are an entity of the same type in the run's training sentences, unseen if they are no entity there, and seen with another type only otherwise; predictions are split the same way, so recall, precision and F1 can be given per kind.
- **Matching criteria** (Segura-Bedmar et al., 2013): span and type, span only, or any overlap, each gold entity matched at most once and an overlap counted in full.

A.5 The dependence model (post hoc)

The units of a corpus share training documents: two MaterioMiner papers' training sets share two papers, two SOFC-Exp folds 27 of their 36 articles, and 5 of the 21 pairs of MuLMS subfields an article. The allowance takes the companion study's model of this dependence (Anonymous, 2026): a correlation of 1/4 between MaterioMiner's papers, 1/5 between SOFC-Exp's folds, and 1/4 between MuLMS subfields whose pools share an article. Each is the correlation the corrected resampled t-test (Nadeau and Bengio, 2003) assumes, the held-out share of the data, 1/K in a design of K folds; the variance of the mean of n units is then $S^2/(1-\bar{\rho}) \cdot (n + \Sigma\rho)/n^2$, summed over the ordered pairs, with n − 1 degrees of freedom. On random splits it takes that test itself, with 96 test and 309 training sentences. Stage 20's protocol fixed its reading in advance, and stage 21's fixes 1/4 between the papers, 1/5 between the folds and the MuLMS pools' layout.

A.6 Stages and protocols

A.7 GLiNER: labels and recipe (stage 21)

- **Model.** GLiNER-medium v2.1 (Zaratiana et al., 2024; Apache-2.0), pinned at one revision for every arm; backbone DeBERTa-v3-base (MIT).
- **Software.** gliner 0.2.30 (Apache-2.0), installed without its dependencies beside the pinned transformers 5.18.0, with sentencepiece 0.2.2, in the stored runs' environment. Each session repeats a stored run, which must be identical.
- **Splits.** Exactly the CRF head's, every unit's test sentences and training and validation positions equal to the stored runs'. MaterioMiner trains on 234–370 sentences and validates on 41–65; SOFC-Exp's folds train on 534–628 and validate on 94–111; MuLMS trains on 340 and validates on 60.
- **Tokens and decoding.** The corpus words are GLiNER's tokens, and it returns word spans. Decoding is flat (no nested or multi-label spans), with a maximum span width of 12 words and a maximum length of 384. Spans become IOB2 tags on the same words, scored by seqeval as every system is.
- **Checks before any number is read.** On gold spans the conversion reproduces the gold tags; the splits equal the stored runs'; the fine-tuned arm at seed 0 on MaterioMiner fine-grained, paper 1, run twice, is identical in every field but its run time.
- **Labels**, each mapping back to its code one to one, given in the corpus's label order, every example carrying the full label set, as the CRF head's closed set does. MaterioMiner fine-grained: the 27 class names, camel case split and lowercased, e.g. "stress amplitude"; the dataset's misspelled "CrackInitation" becomes "crack initiation". MaterioMiner coarse-grained: the codes lowercased, with no zero-shot arm (the protocol took the codes to be undocumented; the release's tag map names every one, found after the

Stage	Machine, seeds	Decided	Decisions (passing)
1	CPU, one run per split	class weights, the CRF head (no protocol)	not counted
2	CPU, one run per split	multi-task training, domain-adaptive pre-training, the seed vote	6 (0)
3	CPU, 3	a mixed-encoder vote, label semantics, self-training, multi-task training, in-domain pre-training	10 (2)
4	label-semantics and self-training votes on CPU, the rest on GPU, 3	the votes of label semantics, self-training and multi-task training; multi-task training with label semantics; self-training with 3 teachers	12 (2)
5	GPU, 10	the re-tests of stages 3 and 4, and the votes	18 (3)
6	GPU, screens 3, decisions 10	screens of DeBERTa-v3-large, document context and multi-task self-training; the last two and their votes	8 (0)
7	GPU, screen 3, decisions 10	a screen of mention replacement; label propagation, recall calibration, coarse-to-fine completion, mention replacement, and their votes	14 (3)
8	GPU, 3	a third paper at equal size	2 (2)
9	GPU, 10	mention replacement on the multi-task model, and its vote	4 (3)
10	GPU, 10	mention replacement on MuLMS, and its votes	2 (1)
11	GPU, 10	names from the other subfields' sentences (MuLMS) and from the fatigue corpus (MaterioMiner)	4 (0)
12	GPU, 10 (MuLMS 7)	names from the held-out topic's corpus, six copies (D1 to D4)	4 (2)
13	GPU, 10	label propagation on held-out documents (13.1a, 13.1b, 13.1d); pooled names (13.2, D5); SOFC-Exp's folds (13.3a, 13.3b; 13.4, D6)	8 (2)
14	GPU, 7 or 10	names against six copies (14.1, D7); the topic's corpus against a mixed one (14.2, D8)	2 (0)
15, 16	—	none	—
17	GPU, 10	names on MaterioMiner's strongest system (D9, D10)	2 (0)
18	GPU, 7	names against a recall-matched bias (D11, D12)	2 (1)
19	GPU	names ranked by a graph network (a screen, failed)	none
20	GPU, 7 or 10	MuLMS's pool by sentence (20.1); a third paper at equal size on fresh seeds (20.2, 20.3)	3 (3)
21	GPU, 10	GLiNER fine-tuned against the CRF head (21.1 to 21.4)	4 (0)
—	GPU, 10	SOFC-Exp's official split and 10-article draws (Table B3)	13 (3)

Table A1. The stages, with the decisions each took and how many pass. The companion study's 3-seed screens are not listed.

run); the exploratory rerun gives each code's class name, the ontology class's label (e.g. "mechanism", "mechanical test parameter"). SOFC-Exp: device, experiment, material, value. MuLMS: the MuLMS paper's names, CITE citation, DEV device, FORM form, INSTRUMENT instrument, MAT material, MEASUREMENT measurement, NUM number, PROPERTY property, RANGE range, SAMPLE sample, TECHNIQUE technique, UNIT unit, VALUE value.

- **Zero-shot arm.** One deterministic run per unit, on MaterioMiner fine-grained (4 papers), MuLMS (7 subfields) and SOFC-Exp (5 folds). The threshold is chosen from 0.3, 0.4, 0.5 and 0.6 by validation micro F1, ties going to the higher threshold; one decode at 0.3 is filtered by score.
- **Fine-tuned arm.** On the CRF head's training sentences, in float32, with deterministic CUDA: AdamW, encoder learning rate 1e-5 and 5e-5 elsewhere, weight decay 0.01, linear warm-up over 10% of the steps; batch 8 (gradient accumulation if memory requires); at most 20 epochs, patience 3, keeping the epoch of best validation micro F1 at threshold 0.5, which is also the test threshold; seed = split seed + offset, as the CRF head's; seed offsets 0–900 on all four settings, all ten seeds on MuLMS since no screen ran.
- **Training details the recipe leaves open** follow the package's training defaults: the focal loss at α -1 and γ 0, i.e. binary cross-entropy, summed; gradient clipping at 1.0; no weight decay on biases or LayerNorm weights; AdamW's default betas and ε; warm-up steps rounded up. The training loop is our own, since the package's trainer needs a library the pinned environment lacks; batches are shuffled per epoch by the CRF head's rule. The exploratory check of the loop runs the package's trainer (with that library added) on the same batches, loss, optimizer groups and learning rates, stopped after each seed's best epoch in our loop's run; only the data order and the random state differ, and a repeat on

the other GPU reproduces it exactly.

- **Detectable differences at 80% power**, from two proxies: a method increment (mention replacement over the CRF head), 1.65 fine-grained, 1.60 coarse-grained, 3.30 on MuLMS and 1.17 on SOFC-Exp; and a different encoder (stage 6's DeBERTa screen), 3.8 fine-grained and 4.8 coarse-grained.
- **Readings.** MaterioMiner reads "above" only if 21.1 and 21.2 both pass, and "below" only if both are below; one granularity alone is reported as that granularity's result and does not count towards "two or more corpora".
- **Disclosed.** A maximum span width of 12 can miss 0 of the gold test entities fine-grained, 1 of 2,194 coarse-grained, 4 of 4,590 on SOFC-Exp and 78 of 31,952 on MuLMS. Under the pinned transformers 5.18, DeBERTa-v3's fast tokenizer maps 32 of MuLMS's 14,423 distinct words (rare non-ASCII characters) to `[UNK]`; MaterioMiner and SOFC-Exp have none.

A.8 Compute

Stages 1 to 3, and stage 4's label-semantics and self-training votes, ran on CPUs; every other decision rests on deterministic GPU runs: T4 GPUs (15 GB) with deterministic CUDA, one frozen commit per batch and a pinned image. The two machines are never compared within a decision, and two probes without decisions trained on CPUs. From stage 5 on, every completed GPU session but two began by repeating an earlier run, and every repeat was identical. Stage 21's trial session repeated a stored run identically, its timed fine-tuning runs took 5 to 8 minutes per unit and the zero-shot arm 0.2 minutes; no result was read before its main batch, which ran over two sessions of about 10.4 and 0.2 hours, the second resuming the first's finished runs.

B Decisions

Stage	Machine, seeds	Protocol	Comparison	Fine-grained Δ [95% CI]	Coarse-grained Δ [95% CI]
3	CPU, 3	random splits	mixed-encoder vote vs seed vote	+0.03 [−1.80, +1.86]	−0.87 [−2.09, +0.35]
3	CPU, 3	unseen papers	label semantics vs CRF head	−0.16 [−2.13, +1.80]	**+1.17 [+0.94, +1.41]**
3	CPU, 3	unseen papers	self-training vs CRF head	+2.91 [−0.73, +6.56]	−0.12 [−1.97, +1.73]
3	CPU, 3	random splits	multi-task vs CRF head	+1.87 [−2.25, +6.00]	**+0.75 [+0.10, +1.39]**
3	CPU, 3	random splits	in-domain pre-training vs CRF head	+0.11 [−2.74, +2.97]	−0.15 [−1.06, +0.76]
4	CPU, 3	unseen papers	label-semantics vote vs seed vote	+0.36 [−2.53, +3.24]	**+1.63 [+0.11, +3.16]**
4	CPU, 3	unseen papers	self-training vote vs seed vote	+3.68 [−1.15, +8.52]	−0.09 [−3.52, +3.35]
4	GPU, 3	random splits	multi-task vote vs seed vote	+1.26 [−1.08, +3.59]	−0.07 [−2.02, +1.87]
4	GPU, 3	random splits	multi-task with label semantics vs CRF head	+0.76 [−0.57, +2.10]	+0.65 [−0.63, +1.93]
4	GPU, 3	unseen papers	multi-task with label semantics vs CRF head	+4.08 [−1.99, +10.14]	−0.01 [−2.64, +2.63]
4	GPU, 3	unseen papers	self-training, 3 teachers, vs CRF head	**+1.07 [+0.33, +1.80]**	+0.85 [−0.58, +2.29]
5	GPU, 10	random splits	multi-task vs CRF head	**+1.83 [+0.32, +3.34]**	+0.71 [−0.15, +1.56]
5	GPU, 10	unseen papers	label semantics vs CRF head	+1.23 [−0.96, +3.42]	−0.36 [−1.66, +0.94]
5	GPU, 10	unseen papers	self-training, 3 teachers, vs CRF head	+1.74 [−1.16, +4.64]	+1.56 [−0.88, +4.00]
5	GPU, 10	random splits	multi-task vote vs seed vote	+1.14 [−0.98, +3.27]	**+0.79 [+0.17, +1.42]**
5	GPU, 10	unseen papers	multi-task vote vs seed vote	**+3.65 [+0.49, +6.82]**	−0.32 [−1.07, +0.44]
5	GPU, 10	random splits	label-semantics vote vs seed vote	−0.79 [−2.02, +0.43]	+0.31 [−1.27, +1.89]
5	GPU, 10	unseen papers	label-semantics vote vs seed vote	+0.57 [−1.46, +2.60]	−0.63 [−1.72, +0.46]
5	GPU, 10	random splits	self-training vote vs seed vote	+0.98 [−0.15, +2.10]	+1.12 [−0.36, +2.60]
5	GPU, 10	unseen papers	self-training vote vs seed vote	+1.12 [−2.71, +4.96]	+0.93 [−1.66, +3.53]
6	GPU, 3	unseen papers	screen: DeBERTa-v3-large vs CRF head	−4.38 [−7.32, −1.45]	−8.01 [−11.72, −4.30]
6	GPU, 3	unseen papers	screen: document context vs CRF head	+2.66 [+0.91, +4.40]	−0.50 [−3.82, +2.83]
6	GPU, 3	unseen papers	screen: multi-task self-training vs CRF head	+6.06 [+1.05, +11.06]	+0.25 [−1.12, +1.62]
6	GPU, 10	unseen papers	document context vs CRF head	+1.14 [−0.94, +3.22]	+0.12 [−2.03, +2.27]
6	GPU, 10	unseen papers	multi-task self-training vs CRF head	+5.47 [−0.74, +11.68]	+0.90 [−1.18, +2.98]
6	GPU, 10	unseen papers	document-context vote vs strongest vote	−1.44 [−3.80, +0.91]	+0.76 [−1.79, +3.31]
6	GPU, 10	unseen papers	multi-task self-training vote vs strongest vote	+1.54 [−1.73, +4.82]	+1.08 [−1.40, +3.57]
7	GPU, 3	unseen papers	screen: mention replacement vs CRF head	+3.25 [−2.33, +8.82]	+0.25 [−2.21, +2.72]
7	GPU, 10	unseen papers	label propagation vs CRF head	+0.30 [−0.35, +0.95]	**+0.12 [+0.01, +0.24]**
7	GPU, 10	unseen papers	label propagation on the strongest vote vs that vote	+0.33 [−0.74, +1.40]	+0.20 [−0.15, +0.55]
7	GPU, 10	unseen papers	recall calibration vs CRF head	−0.59 [−1.75, +0.57]	+0.17 [−0.14, +0.48]
7	GPU, 10	unseen papers	recall-calibration vote vs strongest vote	−3.53 [−8.10, +1.05]	−0.09 [−0.75, +0.58]
7	GPU, 10	unseen papers	coarse-to-fine completion vs CRF head	+0.34 [−0.91, +1.59]	—
7	GPU, 10	unseen papers	coarse-to-fine completion on the multi-task vote vs that vote	+0.12 [−1.76, +2.01]	—
7	GPU, 10	unseen papers	mention replacement vs CRF head	**+2.77 [+1.51, +4.03]**	**+1.28 [+0.06, +2.51]**
7	GPU, 10	unseen papers	mention-replacement vote vs strongest vote	−0.04 [−2.09, +2.01]	+2.26 [−0.12, +4.64]
8	GPU, 3	unseen papers	three papers cut to the size of two vs two papers	**+1.18 [+0.22, +2.15]**	**+3.59 [+1.59, +5.58]**
9	GPU, 10	unseen papers	mention replacement on the multi-task model vs the multi-task model	**+3.01 [+0.42, +5.60]**	**+2.30 [+0.49, +4.10]**
9	GPU, 10	unseen papers	its vote vs strongest vote	**+2.86 [+0.18, +5.54]**	+1.65 [−0.47, +3.78]

Table B1. Decisions and screens on MaterioMiner, stages 3 to 9: the mean paired micro-F1 difference over the 5 random splits or the 4 held-out papers, each side the mean of its seeds; bold, a decision whose interval lies above 0. Screens are not decisions. The strongest vote is the multi-task vote fine-grained and the seed vote coarse-grained. Stage 2's six decisions (one run per split), none passing, are not listed.

Stage	Corpus, units (seeds)	Decision	Comparison	△ micro F1 [95% CI]	Verdict
10	MuLMS, 7 subfields (0–900)	—	mention replacement vs CRF head	+2.10 [−0.31, +4.51]	no
10	MuLMS, 7 (0–900)	—	mention-replacement votes vs CRF votes	+2.71 [+0.22, +5.20]	passes
11	MuLMS, 7 (0–900)	—	names from the other subfields' unlabeled sentences vs CRF head	+2.04 [−0.85, +4.92]	no
11	MuLMS, 7 (0–900)	—	the same vs mention replacement	−0.06 [−1.09, +0.96]	no
11	MaterioMiner, 4 papers (0–900)	—	names from the fatigue corpus vs mention replacement, fine / coarse	+0.23 [−0.69, +1.14] / +0.61 [−0.96, +2.18]	no / no
12	MuLMS, 7 (300–900)	D1	names from the held-out topic's corpus, six copies, vs CRF head	+3.19 [−1.13, +7.51]	no
12	MuLMS, 7 (300–900)	D2	the same vs mention replacement	+0.92 [−1.63, +3.48]	no
12	MaterioMiner, 4 (0–900)	D3 / D4	the MaterioMiner analogue vs CRF head, fine / coarse	+4.27 [+1.39, +7.15] / +1.77 [+0.67, +2.88]	passes / passes
13	MuLMS, 7 (0–900)	13.1a	label propagation vs CRF head	+0.82 [+0.18, +1.46]	passes
13	MuLMS, 7 (0–900)	13.1b	label propagation on mention replacement vs mention replacement	+0.57 [−0.12, +1.25]	no
13	MaterioMiner, 4 (0–900)	13.1d	the same, fine / coarse	+0.19 [−0.34, +0.73] / +0.11 [−0.08, +0.31]	no / no
13	MuLMS and MaterioMiner fine, 11 (0–900)	13.2, D5	names vs CRF head, pooled	+3.58 [+0.95, +6.21], neither corpus below 0	passes
13	SOFC-Exp, 5 folds (0–900)	13.3a	mention replacement vs CRF head	−0.48 [−1.36, +0.39]	no
13	SOFC-Exp, 5 (0–900)	13.3b	mention replacement on the multi-task model vs CRF head	−0.11 [−0.69, +0.46]	no
13	SOFC-Exp, 5 (0–900)	13.4, D6	names vs CRF head	−0.04 [−1.21, +1.14]	no
14	MuLMS and MaterioMiner fine, 11 (300–900 on MuLMS)	14.1, D7	names vs six copies without names, pooled	+0.80 [−0.66, +2.26]	no
14	MuLMS, 7 (300–900)	14.2, D8	the held-out topic's corpus vs a mixed corpus of six other topics	−0.18 [−2.01, +1.65]	no
17	MaterioMiner, 4 (0–900)	D9 / D10	multi-task + MR6 + names vs multi-task + MR6, fine / coarse	+0.96 [−0.01, +1.92] / +0.85 [−0.72, +2.42]	no / no
18	MuLMS, 7 (300–900)	D11	MR6 + names vs MR6 decoded at the bias that matches the names' recall over all test entities: recall on the unseen names the lexicon holds	+11.24 [+3.31, +19.18]	passes as registered (three subfields matched at the grid's end; +6.05 [−3.33, +15.44] on the four matched)
18	MuLMS, 7 (300–900)	D12	the same, micro F1	+3.26 [−0.79, +7.31]	no
20	MuLMS, 7 (0–900)	20.1	the pool drawn by sentence vs by article	+9.51 [+1.12, +17.91]	passes
20	MaterioMiner, 4 (300–900)	20.2 / 20.3	three papers cut to the size of two vs two papers, fine / coarse	+4.03 [+1.06, +7.00] / +3.52 [+1.32, +5.72]	passes / passes

Table B2. Decisions of stages 10 to 20 on held-out documents, the companion study's D1 to D12 among them (MR6: mention replacement with six copies). Each side of a pair is the mean over its seeds.

Comparison	Level	△ micro F1 [95% CI]	Verdict
CRF head vs plain MatSciBERT	entity types	+1.22 [+0.50, +1.93]	passes
test articles cut as the training articles are, the CRF head	entity types	+0.86 [+0.39, +1.33]	passes
the same, plain MatSciBERT	entity types	+1.91 [+1.33, +2.49]	passes
multi-task vs CRF head	slot types	−0.78 [−2.39, +0.84]	no
multi-task vs CRF head	entity types	−0.15 [−1.36, +1.05]	no
mention replacement vs CRF head	entity types	−0.29 [−1.04, +0.47]	no
mention replacement vs CRF head, on 10 of the 26 training articles (5 draws)	entity types	+0.50 [−0.47, +1.46]	no
mention replacement on the multi-task model vs CRF head	slot types	+0.69 [−1.18, +2.57]	no
mention replacement on the multi-task model vs CRF head	entity types	−0.19 [−1.27, +0.88]	no
document context vs CRF head	entity types	+0.58 [−0.26, +1.42]	no
recall calibration vs CRF head	entity types	−0.21 [−0.58, +0.17]	no
label propagation vs CRF head	entity types	−0.42 [−0.61, −0.22]	no

Table B3. Decisions on SOFC-Exp's official split, paired over 10 seeds (intervals conditional on the split's articles), and on its 10-article draws, paired over 5 draws. The rule's second SOFC-Exp decision, on the folds, loses (−0.44 [−0.72, −0.16]).